

Nextflow

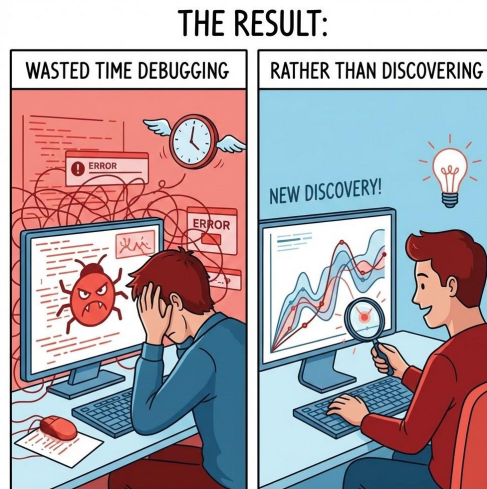
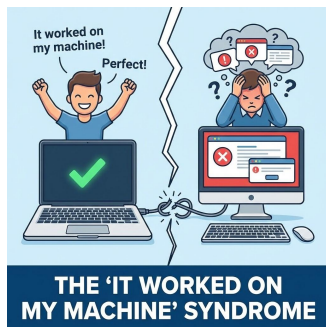


What's Nextflow and why it's useful for you

Some problems that drive us nuts everyday

The "Crisis" of Modern Research

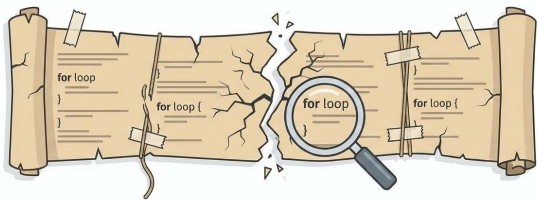
The Problem: Reproducibility



Some problems that drive us nuts everyday

The Daily Struggle

The Problem: Managing Tasks & Repetition



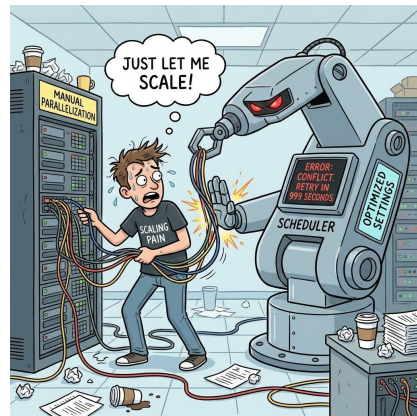
Fragile Scripts

Relying on long Bash scripts with `for` loops that break easily.



The Crash Nightmare

If a 10-hour analysis crashes at hour 9, do you have to restart from hour 0?



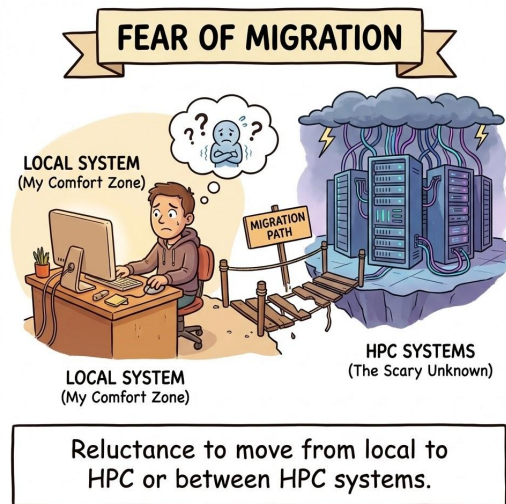
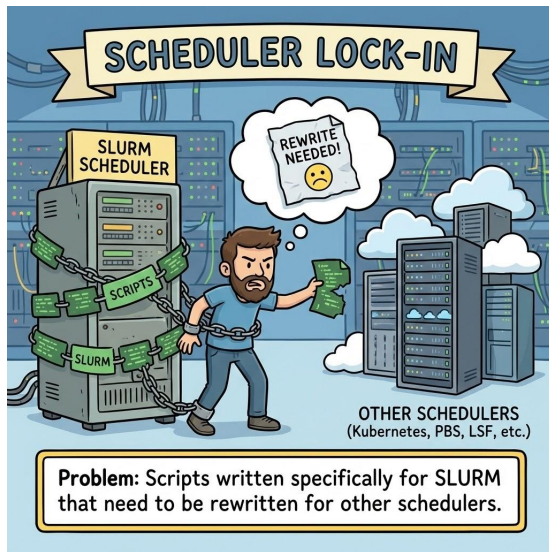
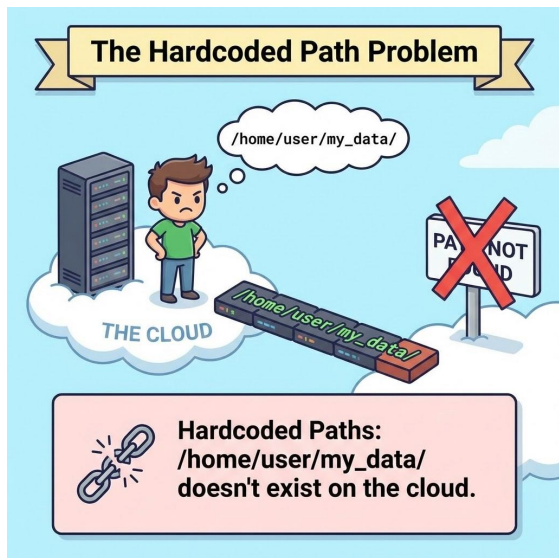
Scaling Pain

Trying to manually parallelize tasks fighting against the scheduler

Some problems that drive us nuts everyday

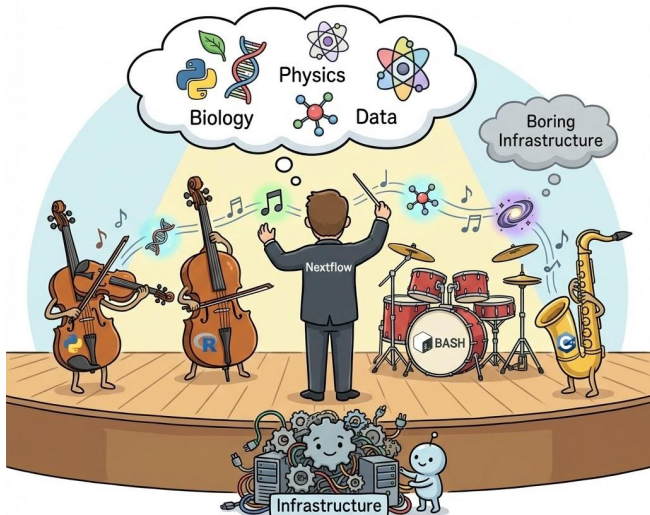
The Infrastructure Barrier

The Problem: The Stress of Portability



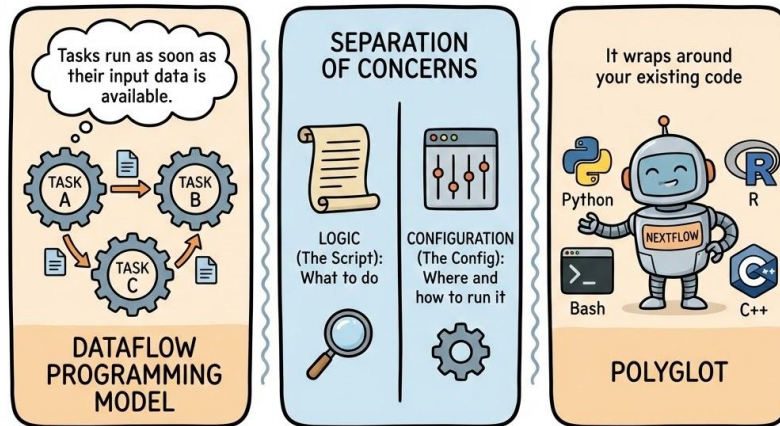
Nextflow helps here

Nextflow: Your Scientific Orchestra Conductor!



Wraps your scripts, manages the **data flow**, and handles the **infrastructure**, so you focus on the **science**!

NEXTFLOW: Workflow Management System & DSL



Summarizing the strengths of Nextflow.

How Nextflow Solves Reproducibility

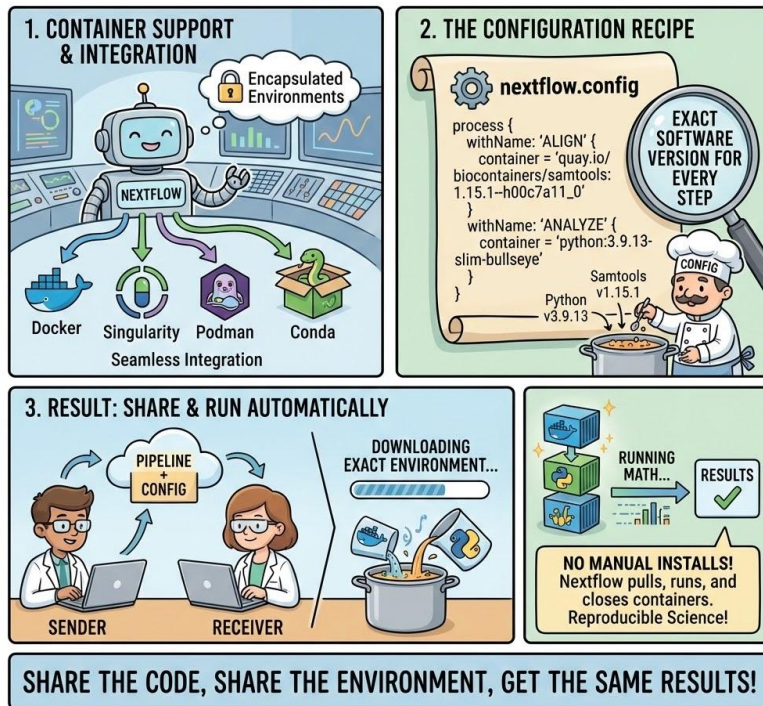
Self-Contained Environments

- **Container Support:** Docker, Singularity, Podman, and Conda.
- **The configuration:** The configuration file defines the exact software version for every step.

You share the pipeline, and Nextflow automatically downloads the exact environment required to run it. Nextflow pulls the container, runs the math, and closes it down.

Every time you run your analysis you get consistent results!

NEXTFLOW: SOLVING REPRODUCIBILITY WITH CONTAINERS



How Nextflow Solves Task Management

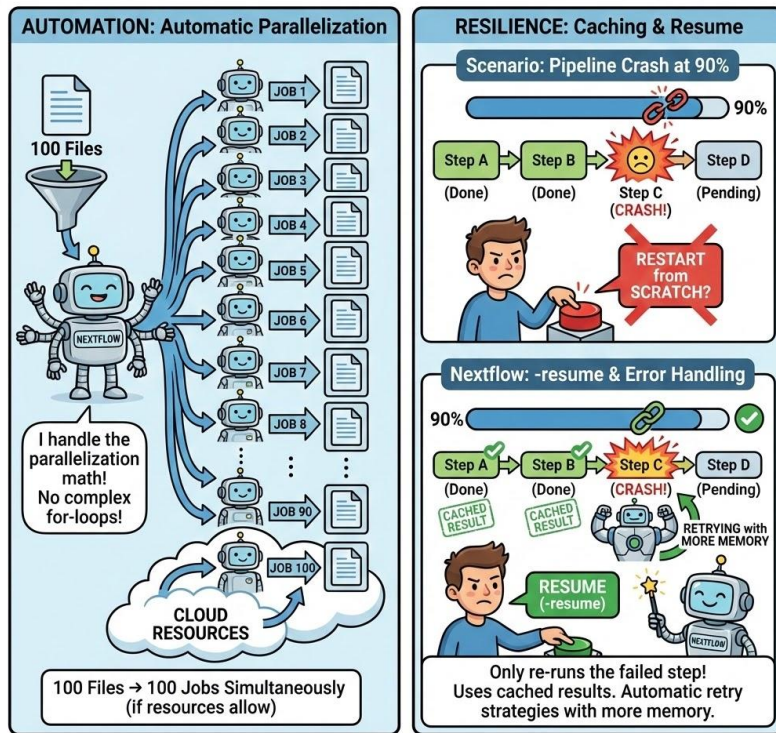
Automation & Resilience

- **Automatic Parallelization:** Nextflow automatically forks tasks.
- **Resume Capability:** Only re-runs from the failed step.
- **Error Handling:** Strategies to automatically retry failed jobs.

No more restarting from scratch.

No more complex for-loops and scheduler dependencies.

NEXTFLOW: AUTOMATION & RESILIENCE

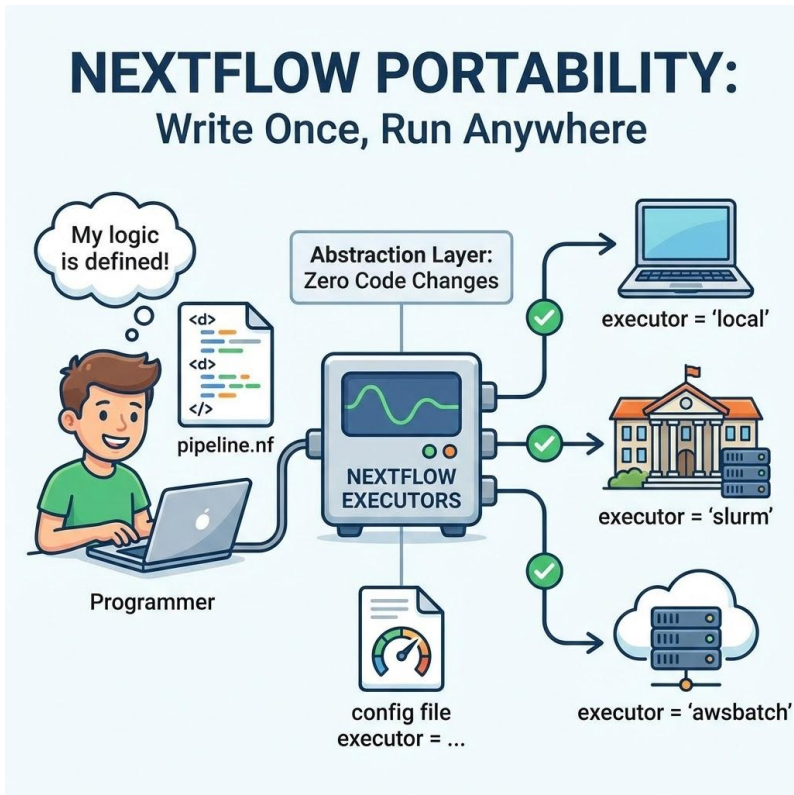


How Nextflow Solves Portability

Write Once, Run Anywhere

- **Executors:** An abstraction layer between your code and the hardware.
- **Switching Infrastructure:** As easy as changing executor configuration
- **Zero Code Changes:** The pipeline script remains exactly the same.

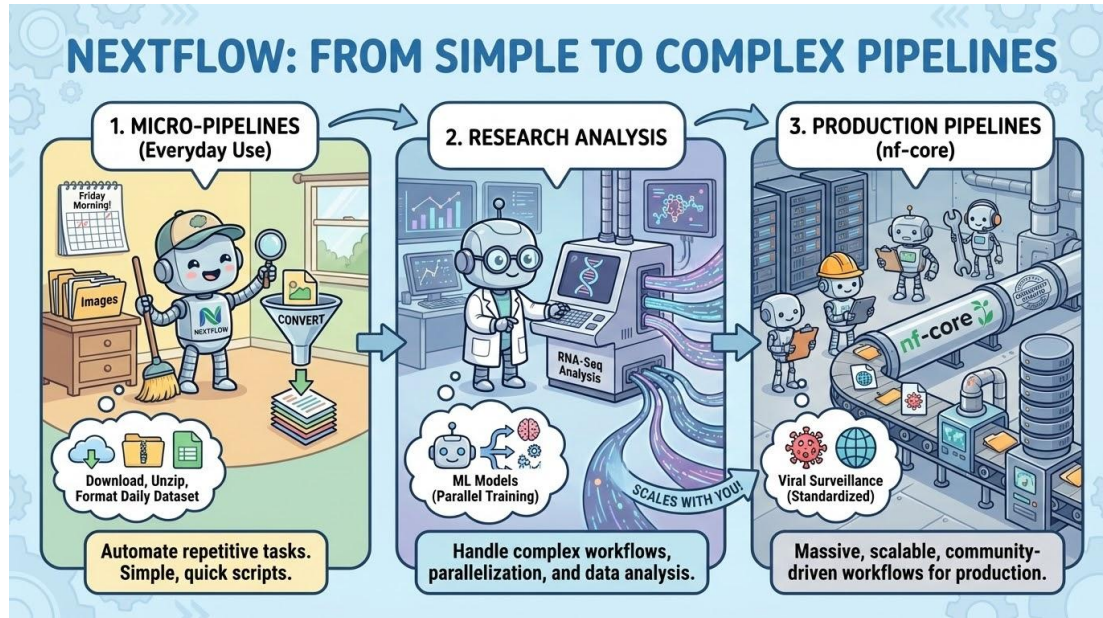
You define the logic once. You change few lines in a config file and Nextflow translates your needs to the cluster's language.



It doesn't need to be complex

From Simple to Complex

Nextflow is famous for massive genomics pipelines, but that's not its only use. It is equally powerful for automating that annoyingly repetitive task you do every morning.



Take-away messages

Nextflow as an investment in your future sanity

Why Nextflow is for YOU (Every Day) → A tool to organize your data handling and processing.

- **You don't need a massive pipeline.** Even a 3-step process benefits from caching and error handling.
- **Peace of Mind:** Knowing your work is reproducible and safe from crashes.
- **Modularity:** Build small blocks of code you can reuse later.
- **Future Proofing:** If your data grows 10x or you have to re-run with many different configurations, your Nextflow script handles it without rewriting.

